CASE STUDY: RANSOMWARE ATTACK ON A COLLEGE NETWORK (FICTIONAL SCENARIO)

Ransomware is a type of cybercrime in which attackers use malicious software to block access to a computer system or encrypt important data, demanding payment (usually in cryptocurrency) to restore access. It has become increasingly common in educational institutions due to their large user base and often weaker cybersecurity systems.

In this case, a mid-sized college in India experienced a ransomware attack during the examination period. The attack began with a phishing email sent to multiple faculty members. Step 1: The email appeared to be from the college administration and contained an attachment labeled “Updated Exam Schedule.” Step 2: A faculty member downloaded and opened the attachment, which silently installed malware on their system. Step 3: The malware spread through the college’s internal network, exploiting weak passwords and unpatched systems. Step 4: Within hours, critical files—including student records, exam data, and administrative documents—were encrypted. Step 5: A message appeared on all affected systems demanding payment in cryptocurrency within 72 hours, threatening permanent data loss.

The primary targets in such attacks are institutions like colleges, hospitals, and small businesses that rely heavily on continuous access to digital systems but may lack strong cybersecurity infrastructure. Individuals within these organizations—such as staff or students—are often the entry point due to human error, such as clicking on malicious links or downloading unsafe files.

The consequences of ransomware attacks are severe. In this case, the college had to postpone exams, causing academic disruption and stress for students. Sensitive student data was at risk of being leaked, raising

privacy concerns. Financial losses included potential ransom payment, system recovery costs, and investment in stronger security measures afterward. Additionally, the institution’s reputation suffered, as students and parents lost trust in its ability to protect data.

This case highlights the importance of cybersecurity awareness, regular system updates, data backups, and training users to recognize phishing attempts to prevent such attacks.